Daniel Amaya Zabala

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Education

Universidad de Antioquia Medellín, CO
BSc in Mathematics (Expected Graduation: 2026) GPA: 4.05 / 5.0
Relevant Coursework: Probability, Analysis, Linear Algebra, Topology, abstract algebra, differential geometry.
Grants and awards:

- Top Student of the 2023-2 and 2024-2 Semesters.
- ELAP Scholarship Recipient (Saint Mary's University, Canada, 2024-2).
- Samsung Innovation Campus Scholarship Winner (2024-2).

Study Abroad Saint Mary's University (SMU) GPA: 4.15 / 4.30 Halifax, CA
Study abroad coursework in Artificial Intelligence, Introduction to Numerical Analysis, Complex networks, and French.
August 2024 – December 2024

Work Experience

Summer Research Intern — Simon Fraser University, Vancouver, Canada · Jun 2026 – Sep 2026
Mitacs Globalink Research Program · Supervised by Prof. Ben Adcock

- Modelled and generalized Bayesian priors to quantify the complexity of data sets through the covering number of their underlying probability distributions.
- Ran experiments in PyTorch implementing greedy-cover algorithms, searching for numerics sharp enough to approach the theoretical bounds we had estimated; tested primarily on Fashion-MNIST and CelebA.
- Scoped an open research question with no predefined solution path, distilling a dense body of prior literature into a working framework and isolating which assumptions actually drove the results.
- Reported findings weekly to a senior supervisor and documented results reproducibly, so that collaborators could verify the arguments and build on them. Contributed proofs to a paper in preparation.

Teaching Assistant — Universidad de Antioquia, Medellín, Colombia · Oct 2025 – Jun 2026
Merit-based academic stipend · 10 hours per week

- Taught abstract quantitative material, mainly linear algebra and analytic geometry, to undergraduates of varied backgrounds, calibrating the level of detail to the audience.
- Diagnosed the points where students consistently got stuck and rebuilt study materials to target those specific gaps.
- Designed and led review sessions connecting theory to concrete application, sustaining a fixed weekly commitment alongside full-time coursework.
- Fielded questions live and reformulated explanations on the spot.

Projects & Research

Final Capstone Project – Samsung Innovation Campus (May 2025 – Jul 2025) | GitHub Repository

- Collaborated with a multidisciplinary team to meet the certification criteria set by Samsung Innovation Campus.
- Developed an AI-powered diagnostic system to classify chest X-ray (CXR) images into **bacterial pneumonia**, **viral pneumonia**, or **normal** conditions.
- Applied **Generative Adversarial Networks (GANs)** to address severe class imbalance by oversampling the underrepresented 'normal' and 'viral' categories.
- Fine-tuned a **VGG16** architecture for multi-class classification with key performance metrics:
  - Accuracy: **87.18%**
  - Macro Precision: **86.57%**, Macro Recall: **86.66%**, Macro F1-Score: **86.27%**

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CART VS ID3 Model for Over-the-counter Medical Recommendations - SMU (*Supervised by Mengjun Hu, PhD*)

- I implemented and compared decision tree algorithms to optimize the classification of pharmaceutical data.
- Implemented Gini impurity calculations and conducted model evaluations to explore potential accuracy improvements.

Exploring Influential Nodes Using Global and Local Information - SMU (*Supervised by Somayeh Kafaie, PhD.)*

- Investigated centrality measures to identify key nodes in complex networks.
- Compared the effectiveness of local (e.g., degree centrality) and global (e.g., betweenness centrality) metrics in assessing node influence.

Directed Reading in Topological Data Analysis (TDA) – Universidad Nacional Autónoma de México (*Supervised by Carlos Segovia, PhD*) · Feb 2024 – Jun 2024

- We studied Morse theory for cell complexes to understand the ideas behind TDA, and we examined applications using the Topology Toolkit software based on the article, "*Morse Theory for Cell Complexes*" by Robin Forman.

Leadership & Activities

Intercultural Buddy / Student Mentor — Universidad de Antioquia, Medellín, Colombia · Jul 2023 – Aug 2024
Programa Parceros (Gestores para la Ciudadanía Intercultural y Global)

- Led initiatives to support international students in adapting to new academic and cultural environments.
- Organized events fostering cultural exchange and global citizenship.
- Enhanced leadership, teamwork, and communication skills by mentoring students and resolving challenges.

Skills & Interests

- **Areas of Expertise:** Python, PyTorch, LaTeX, GitHub; linear algebra, probability, analysis, partial differential equations, Bayesian statistics, inverse problems, topology, data analysis.
- **Languages:** Spanish (Native), English (Full professional proficiency, C1), French (Elementary).
- **Interests:** AI research, problem-solving, large language models, AI agents, and machine learning theory.